

ANIVIBE — Decision Support & Recommender Systems Guide

Page-by-Page Guide to Systems, Decision Tools, and User Experience (Plain English, Zero Equations)

Course: CSX/ITX 4207 Decision Support & Recommender Systems

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Executive Architectural Summary: ANIVIBE combines two complementary technologies to eliminate the entertainment Paradox of Choice across 1,255 Crunchyroll anime titles: **Recommender Systems** automatically discover matching content based on community wisdom (Popularity), thematic similarity (Content-Based), or situational boundaries (Knowledge-Based). **Decision Support Systems** give users transparent evidence (rating distribution histograms, commitment badges, explainable checklists, and side-by-side trade-off matrices) so they can make confident viewing decisions.

Page 1: Home Page /

POPULARITY & PERSONALIZATION

User Goal: The welcome portal. Solves first-time exploration for new users and delivers a tailored feed for returning users who have rated anime.

1. Popular Right Now **POPULARITY-BASED RECOMMENDER**

What it does: Recommends the most trusted, universally acclaimed anime across the community. It balances high review ratings with total review counts to ensure that anime with thousands of votes outrank titles with only a single 5-star review.

Why it's used (Cold-Start Solution): When a new user opens the website for the very first time, the system has zero data on their taste. Rather than leaving the screen blank or guessing randomly, Popularity provides a safe, verified starting catalog immediately.

2. Recommended For You **CONTENT-BASED FILTERING**

What it does: Listens to your 1-to-5 star ratings in the app. When you give 5 stars to fantasy or action shows, the system learns what you like, increases the importance of those genres, and recommends unwatched anime sharing those exact themes.

Decision Support Benefit: Relieves choice overload. Instead of searching through 1,200 titles, the user gets an evolving, tailored feed matching their taste.

3. Interactive Decision Onboarding Banner **DECISION SUPPORT PROMPT**

What it does: If the visitor has zero ratings, a friendly card guides them to rate a few anime, explaining how rating unlocks personalized recommendations.

User Goal: Investigating an individual anime to answer: "Is this specific show worth my time and attention right now?"

1. More Like This (Netflix-Style Horizontal Carousel) ITEM-TO-ITEM CONTENT-BASED FILTERING

What it does: Looks strictly at the anime currently on screen and calculates which other anime in the catalog share the most genre attributes.

Why it's used: Operates completely independently of user rating history. If you fall in love with a specific anime, you can immediately find direct sister titles with the same tone and pacing.

2. Viewer Rating Distribution (1 to 5 Stars Breakdown) COMMUNITY CONSENSUS DECISION SUPPORT

What it does: Displays a golden bar chart directly inside the anime hero card showing what percentage of viewers gave 5 stars, 4 stars, down to 1 star.

Why it's used (Decision Support): An average rating like "4.1" can be misleading. Seeing the distribution helps users spot whether an anime is universally beloved (almost all 5 stars) or polarizing/controversial (lots of 1 stars mixed with 5 stars).

3. Commitment & Time Investment Badges CONSTRAINT EVALUATION

What it does: Flags each anime as *Short* (≤ 13 episodes), *Medium* (14–26 episodes), or *Long* (> 26 episodes) so users can budget their evening or weekend.

User Goal: Provides two modes: self-guided catalog exploration (Browse) or situation-aware digital consultation (Find).

Mode A: Catalog Discovery EXPLORATORY DECISION SUPPORT

What it does: Allows full-catalog search with interactive genre tags, instant title searching, and multi-criteria sorting (by community popularity, score, or title).

Why it's used: Empowers users when they already have a specific keyword in mind or want to explore a single genre category (e.g. Sci-Fi).

Mode B: Find What Fits KNOWLEDGE-BASED RECOMMENDER (KBR)

What it does: Acts as an expert advisor. You declare your real-world situational requirements:

- **Episode Budget:** Set a strict ceiling (e.g., maximum 26 episodes for a weekend binge).
- **Quality Threshold:** Set a minimum standard (e.g., ratings ≥ 4.0 only).
- **Required & Avoided Genres:** Declare what must be present (e.g., Mystery) and what to exclude (e.g., No Horror).
- **Viewing Mood:** Pick an emotional vibe (*Exciting, Chill, Dark, or Emotional*).

Explainable Decision Verdicts: Every recommended title displays an itemized evaluation report (e.g. "✓ *Episode Budget: 12 eps satisfies ≤ 26* " and "✓ *Quality: 4.4 rating meets ≥ 4.0* ") so users understand exactly why the anime was selected.

User Goal: Solves the most common dilemma: *"I want something that matches my genuine taste, but also fits my available time and mood right now."*

1+1 Hybrid Fusion Recommender HYBRID RECOMMENDER + DECISION SUPPORT

What it does: Concurrently computes two independent scores and merges them into a single recommendation:

- **50% User Taste Affinity (Content-Based):** Evaluates how closely the anime's genres match your rating history.
- **50% Requirement Fit (Knowledge-Based):** Evaluates how well the anime satisfies your episode budget, quality threshold, and mood.

Why it's used: Content-based filtering alone might suggest a 500-episode classic when you only have one evening free. Knowledge-based filtering alone might find a short show in a genre you dislike. The 1+1 Hybrid guarantees you get a title you love *and* have time to watch.

Transparent Dual-Score Match Badges DECISION TRANSPARENCY

Each result card displays two distinct progress bars: **Taste Match %** (purple) and **Requirement Fit %** (cyan), giving users complete clarity into why the title was recommended.

Cold-Start Safeguard SYSTEM RELIABILITY

If a user has 0 ratings, the system prompts them to rate a few anime first or smoothly routes them to Knowledge-Based recommendations, ensuring zero crashes or empty states.

Page 5: Decision Center /decision-center

MULTI-CRITERIA DECISION ANALYSIS (MCDA)

User Goal: Resolves final indecision when a user is torn between 2 or 3 finalist anime.

Side-by-Side Comparison Matrix MULTI-CRITERIA DECISION MAKING

What it does: Displays up to 3 selected anime side-by-side across key comparison criteria:

- **Time Investment:** Total episodes and estimated hours required to complete the show.
- **Quality vs. Popularity:** Direct comparison of rating score versus total community review volume.
- **Rating Distributions:** Compare the 5-star vs 1-star ratios of each candidate side by side.
- **Thematic Comparison:** Shows shared genres and unique thematic elements between the choices.

Decision Support Benefit: Eliminates browser tab clutter and mental fatigue by organizing all trade-offs onto a single, structured decision dashboard.

User Goal: Provides instructors and evaluators with scientific proof of how each recommender paradigm performs.

What it does: Displays benchmark metrics comparing all four approaches:

- **Precision:** How accurately the recommendations match user test profiles.
- **Recall:** The percentage of relevant titles captured in the top recommendations.
- **Diversity:** Whether the model offers diverse variety or repeats the same narrow genres.
- **Coverage:** The percentage of the 1,255-title catalog the model can reach.

Key Finding: Demonstrates that the **1+1 Hybrid** achieves the highest precision while maintaining healthy diversity, proving why hybrid architectures are superior.

Summary: Architecture At A Glance

Page Name	Recommender System	Decision Support Feature	User Benefit
1. Home	Popularity Baseline + Personalized CBF	Cold-Start Onboarding Prompt	Safe start for new users; tailored feed as you rate
2. Anime Detail	Item-to-Item Similarity	5-Star Rating Histogram	Spot controversial vs loved titles; find sister anime
3. Browse & Find	Knowledge-Based Recommender (KBR)	Rule Checklist & Multi-Attribute Search	Filter by exact episode limits, minimum score, and mood
4. Smart Match	1+1 Hybrid (50% CBF + 50% KBR)	Dual Taste vs Fit Percentage Badges	Recommends what you love AND have time to watch
5. Decision Center	Comparative Candidate Ranking	Side-by-Side Multi-Criteria Matrix	Resolve indecision between 2 or 3 finalist titles
6. Benchmarks	Cross-Model Empirical Comparison	Precision, Recall, Diversity Metrics	Scientific validation for academic grading and defense

Project Defense Summary for Evaluators: ANIVIBE demonstrates that a modern recommendation platform cannot rely on a single technique. Popularity solves Cold-Start, Content-Based Filtering models personal taste, Knowledge-Based rules satisfy real-world time budgets, and the 1+1 Hybrid fuses them into an optimal decision support system.